



Department of Engineering Technology

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Introduction to Web Technology Lab

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Title: Midterm exam

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Remarks (if any):

Name & Signature of faculty:

1. Introduction

This lab project is a **Student Task Manager Web Application** developed using HTML, CSS, and JavaScript.

The purpose of this application is to allow students to manage their tasks by adding, viewing, filtering, sorting, and loading tasks from a fake API.

2. Objective

The objective of this lab is to:

- Understand frontend development using HTML, CSS, and JavaScript
- Practice DOM manipulation
- Use ES6 features
- Implement filtering and sorting
- Work with fake API using Fetch method

3. Tools & Technologies Used

- HTML5 (Structure)
- CSS3 (Styling & Layout)
- JavaScript (Logic & Functionality)
- Fake API (jsonplaceholder.typicode.com)

4. Application Development Process

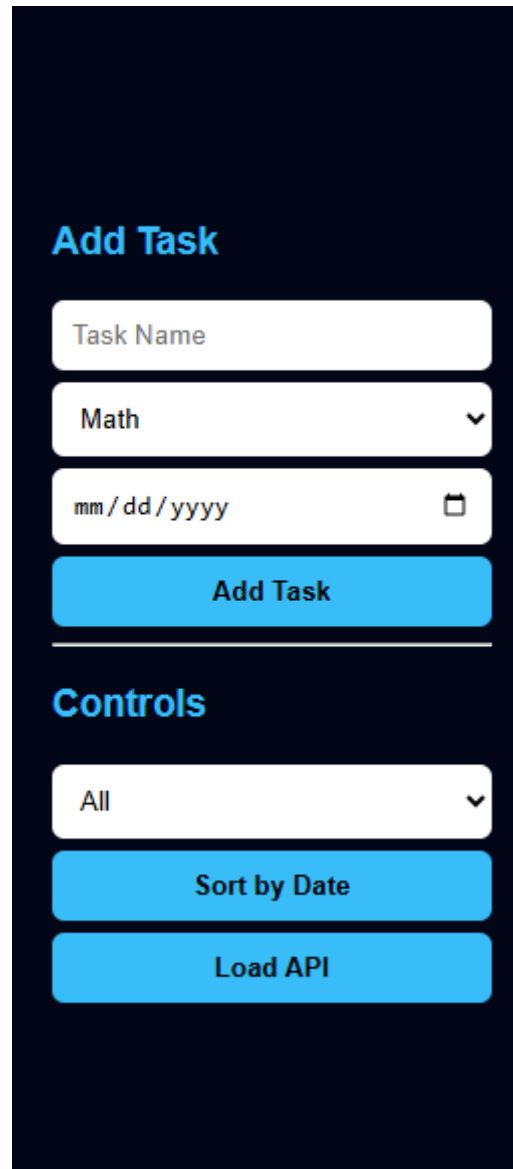
Step 1: HTML Structure

A structured webpage was created containing:

- Task input form (Task Name, Subject, Due Date)

- Sidebar controls
- Task display section

Screenshot 1: Input Form



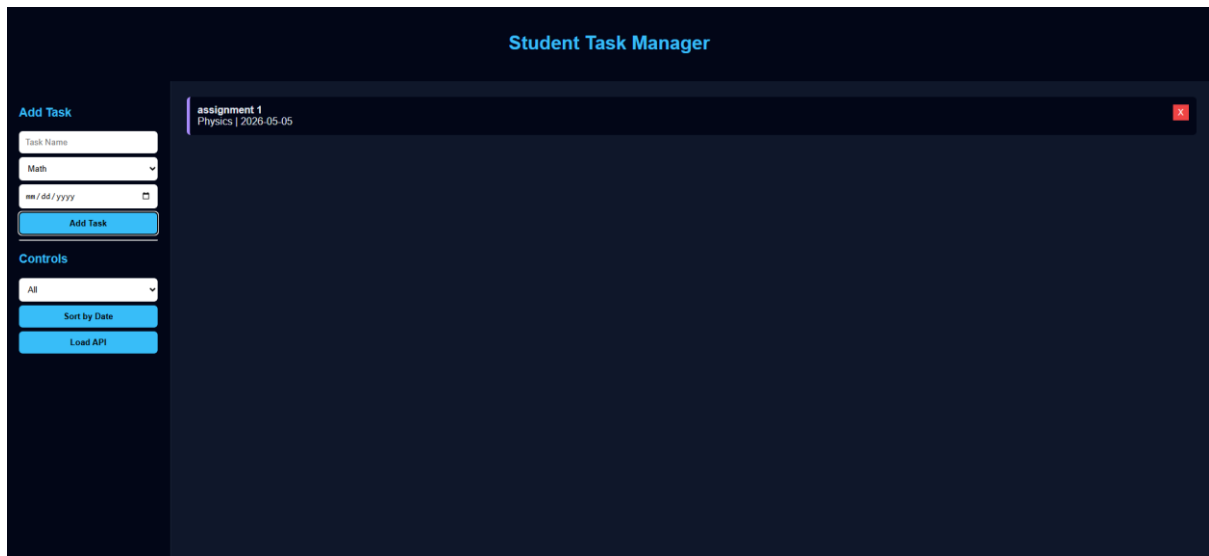
The screenshot shows a mobile application interface with a dark blue background. At the top, there is a section titled "Add Task" in a light blue font. Below this title, there are three input fields: a text field labeled "Task Name", a dropdown menu currently showing "Math", and a date field with the placeholder "mm/dd/yyyy" and a calendar icon. Below these fields is a large blue button labeled "Add Task". A horizontal line separates this section from the "Controls" section below. The "Controls" section also has a light blue title and contains a dropdown menu currently showing "All", followed by two more blue buttons labeled "Sort by Date" and "Load API".

Step 2: CSS Styling

CSS was applied to:

- Create dark theme UI
- Use Flexbox for layout
- Make sidebar + main section structure
- Add responsiveness using media queries
- Improve readability and spacing

Screenshot 2: UI Layout



Step 3: JavaScript Functionality

JavaScript was used to:

- Store tasks in an array
- Add tasks using form submission
- Display tasks dynamically using DOM
- Delete tasks
- Reset input fields after adding task

ES6 Features used:

- Arrow functions
- Template literals
- Array methods (filter, map, sort)

Step 4: API Integration

A fake API was used:

<https://jsonplaceholder.typicode.com/todos>

- Data is fetched using Fetch API

- Tasks are dynamically added to the list

Screenshot 3: API Data Output



Step 5: Filtering & Sorting

- Tasks can be filtered by subject (Math, CS, Physics)
- Tasks can be sorted by due date

Screenshot 4: Filter & Sort Result



5. Code Overview

The system includes:

- Task array storage
- DOM manipulation
- Event handling
- API fetching
- Filtering and sorting logic

6. Output / Results

The application successfully:

- Adds tasks dynamically
- Displays tasks in UI
- Filters tasks by subject
- Sorts tasks by due date
- Loads API data correctly

7. Conclusion

In this lab, I learned how to:

- Build a complete web application using HTML, CSS, and JavaScript
- Use DOM manipulation for dynamic UI updates
- Integrate fake APIs using Fetch method
- Apply ES6 features in real projects
- Implement filtering and sorting logic

This project improved my understanding of frontend development and client-side programming.